H9

```r
library(tidyverse)
```

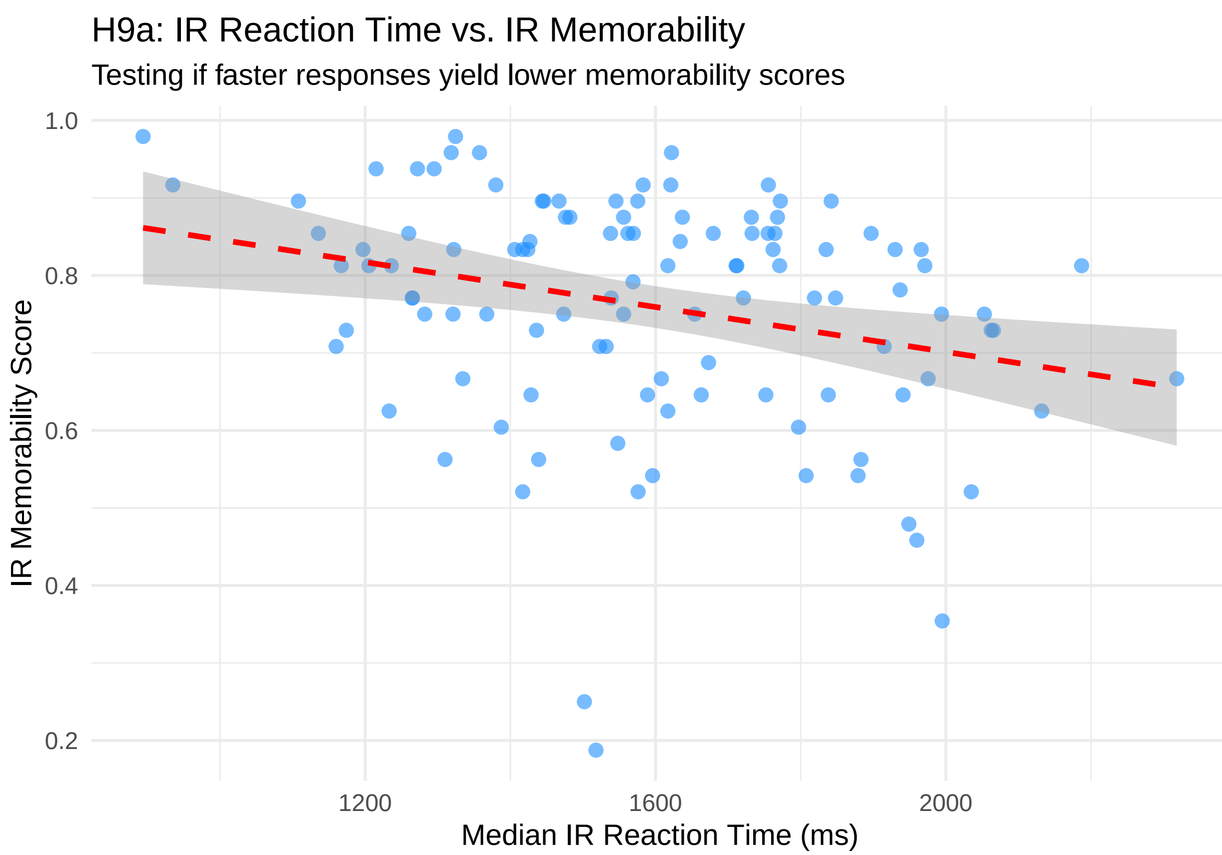
```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
raw_dataset <- read_csv(
  "sentence_memorability_data.csv",
  na = c("", "NA", "N/A"),
  show_col_types = FALSE
)

h9_data <- raw_dataset %>%
  # Make sure RTs are treated as numbers
  mutate(
    Reaction_time_IR = suppressWarnings(as.numeric(Reaction_time_IR)),
    Reaction_time_WR = suppressWarnings(as.numeric(Reaction_time_WR)),
    Accuracy.IR = suppressWarnings(as.numeric(Accuracy.IR)),
    Accuracy.WR = suppressWarnings(as.numeric(Accuracy.WR)),
    prev_IR = lag(Accuracy.IR)
  ) %>%
  group_by(participant_ID) %>%
  summarise(
    # Calculate median reaction times (ignores outliers)
    Median_RT_IR = median(Reaction_time_IR, na.rm = TRUE),
    Median_RT_WR = median(Reaction_time_WR, na.rm = TRUE),

    Mean_RT_IR = mean(Reaction_time_IR, na.rm = TRUE),
    Mean_RT_WR = mean(Reaction_time_WR, na.rm = TRUE),

    # IR components
    Total_Repeats = sum(isRepeat == TRUE & Event == "Sentence shown", na.rm = TRUE),
    Total_Denom = sum(is.na(isRepeat) & Event == "Sentence shown", na.rm = TRUE),
    Hits = sum(Event == "IR pressed" & isRepeat == TRUE & Accuracy.IR == 1, na.rm = TRUE),
    False_Alarms = sum(Event == "IR pressed" & is.na(isRepeat), na.rm = TRUE),

    # IR metrics
    Hit_Rate = if_else(Total_Repeats > 0, Hits / Total_Repeats, 0),
    False_Alarm_Rate = if_else(Total_Denom > 0, False_Alarms / Total_Denom, 0),
    IR_Memorability = Hit_Rate - False_Alarm_Rate,
```

```
    # WR components
    WR_correct = sum(Event == "WR pressed" & Accuracy.WR == 1 & prev_IR == 1, na.rm = TRUE),
    WR_total   = sum(Event == "WR pressed" & prev_IR == 1, na.rm = TRUE),

    # WR metric
    WR_Accuracy = if_else(WR_total > 0, WR_correct / WR_total, 0),
    .groups = "drop"
  ) %>%
  # Drop any participants missing RTs or scores to ensure a clean correlation
  drop_na(Median_RT_IR, Median_RT_WR, IR_Memorability, WR_Accuracy)
```

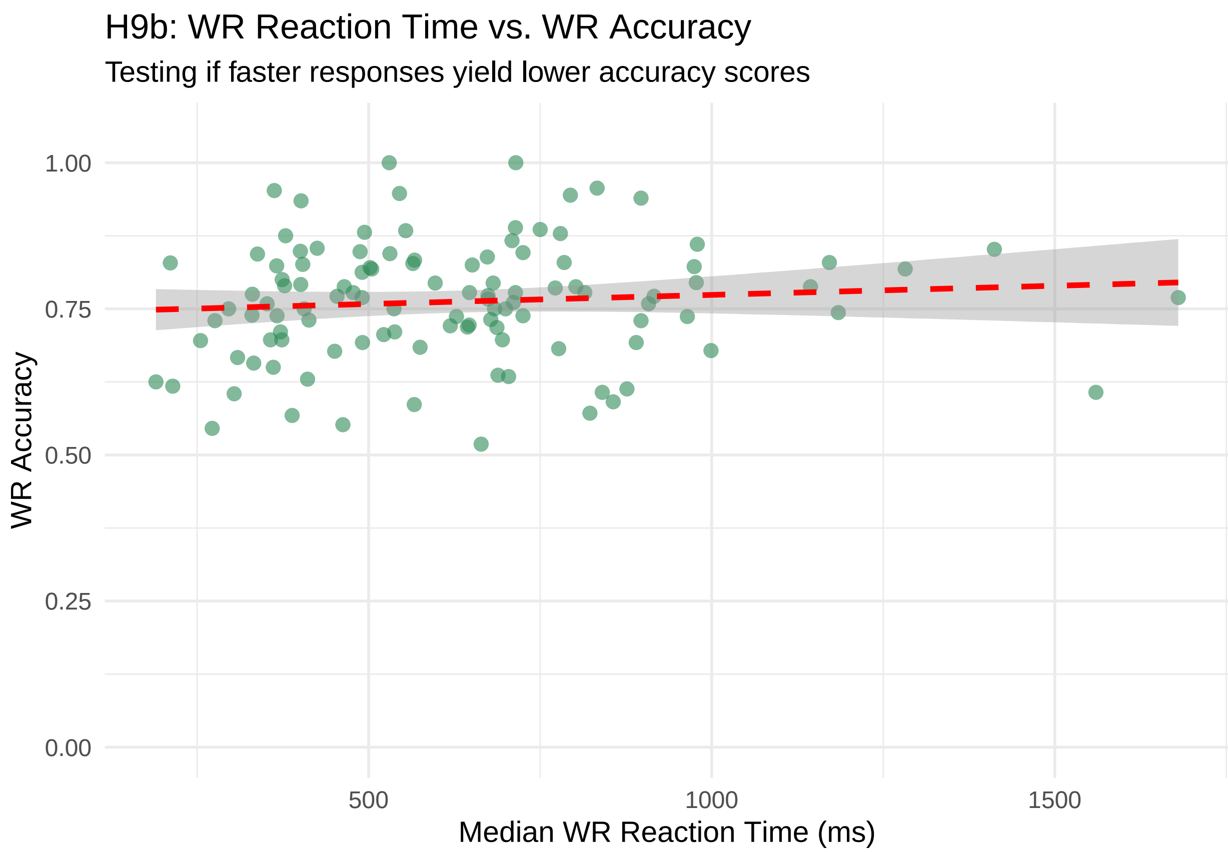
```
library(ggplot2)

# Graph H9a: IR Reaction Time vs. IR Memorability
ggplot(h9_data, aes(x = Median_RT_IR, y = IR_Memorability)) +
  geom_point(alpha = 0.6, color = "dodgerblue", size = 2) +
  geom_smooth(method = "lm", color = "red", linetype = "dashed", se = TRUE) +
  labs(title = "H9a: IR Reaction Time vs. IR Memorability",
       subtitle = "Testing if faster responses yield lower memorability scores",
       x = "Median IR Reaction Time (ms)",
       y = "IR Memorability Score") +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

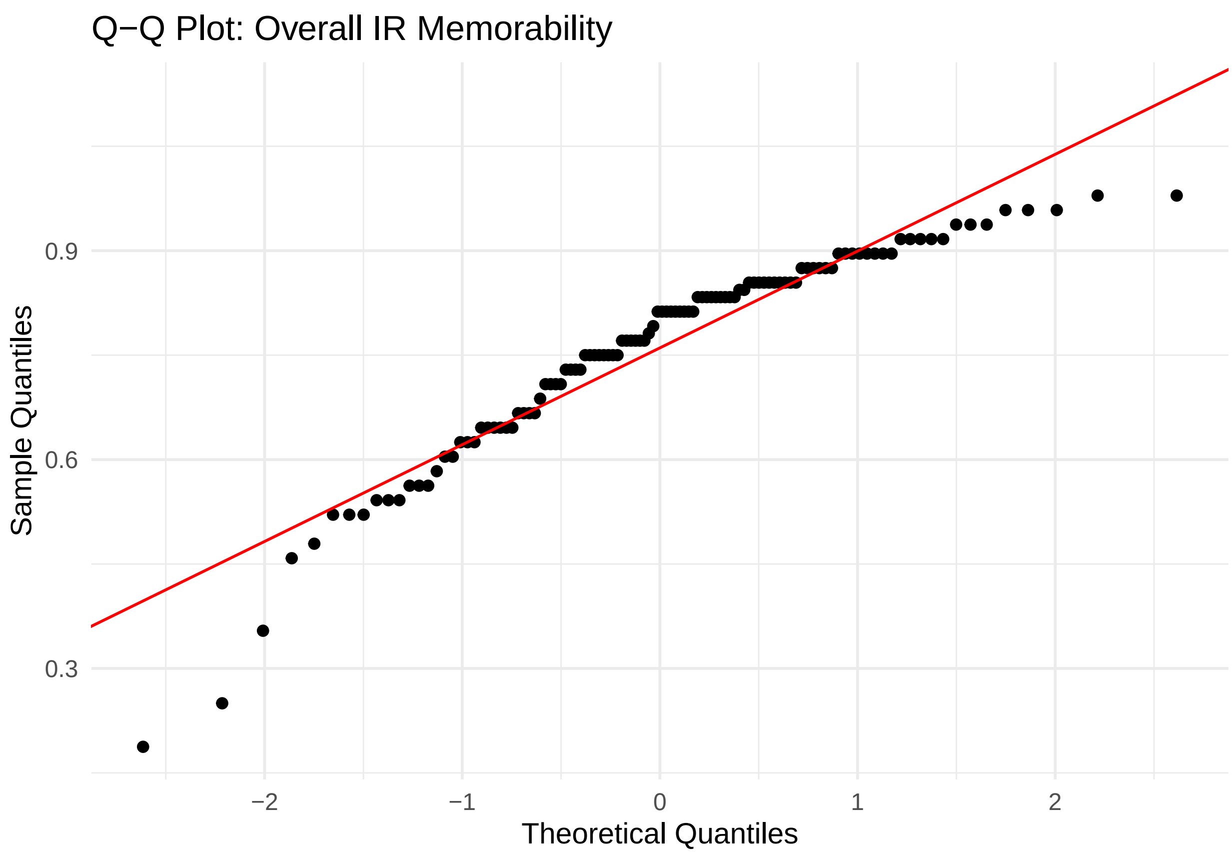


```r
# Graph H9b: WR Reaction Time vs. WR Accuracy
ggplot(h9_data, aes(x = Median_RT_WR, y = WR_Accuracy)) +
  geom_point(alpha = 0.6, color = "seagreen", size = 2) +
  geom_smooth(method = "lm", color = "red", linetype = "dashed", se = TRUE) +
  scale_y_continuous(limits = c(0, 1.05)) +
  labs(title = "H9b: WR Reaction Time vs. WR Accuracy",
       subtitle = "Testing if faster responses yield lower accuracy scores",
       x = "Median WR Reaction Time (ms)",
       y = "WR Accuracy") +
  theme_minimal()
```

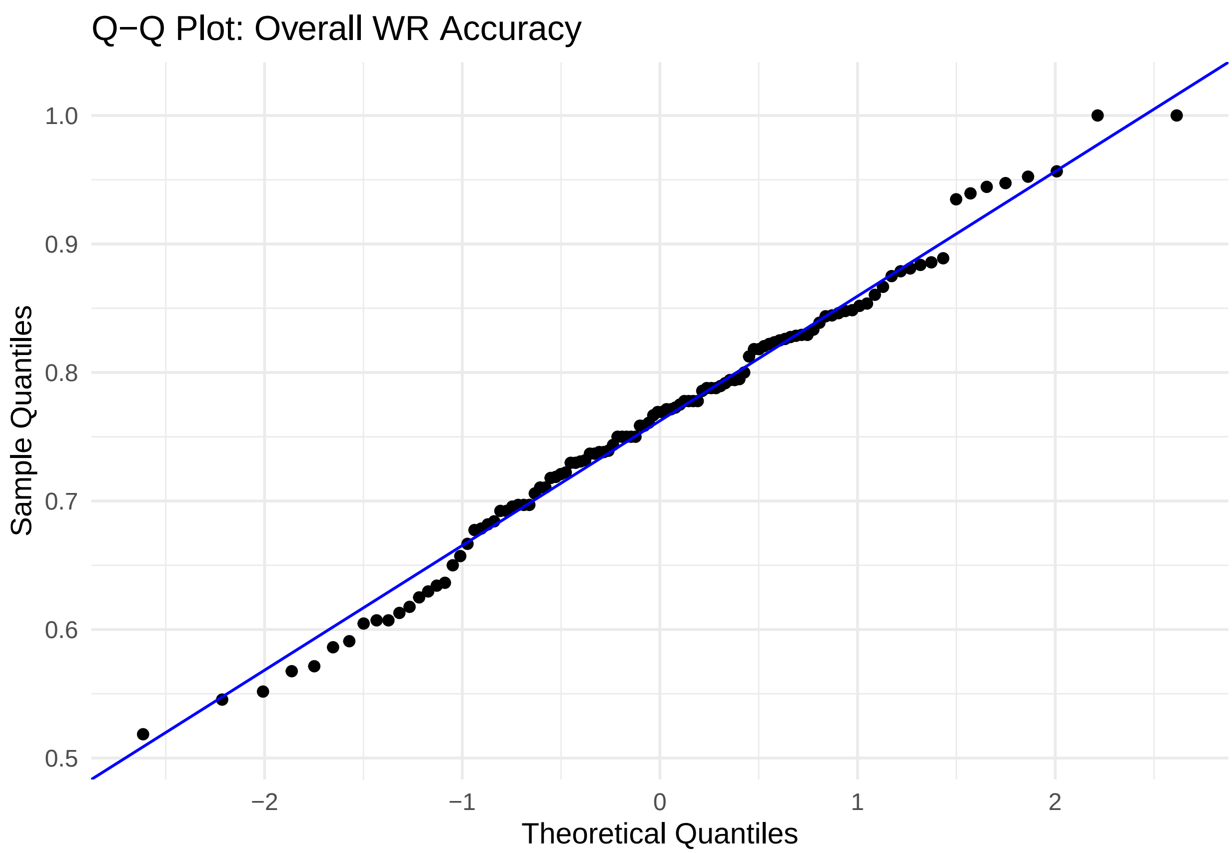
```
## `geom_smooth()` using formula = 'y ~ x'
```



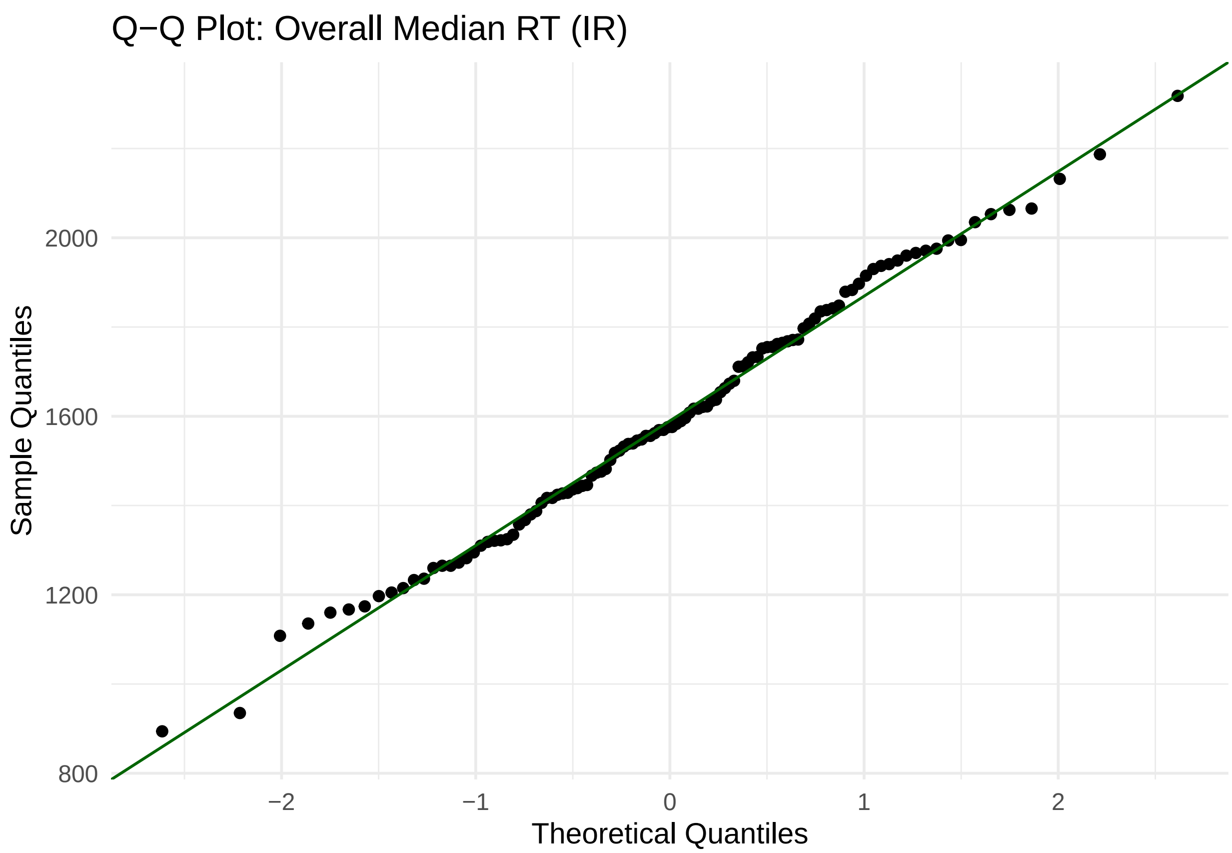

```r
# Q-Q Plot for Overall IR Memorability
ggplot(h9_data, aes(sample = IR_Memorability)) +
  stat_qq() +
  stat_qq_line(color = "red") +
  labs(title = "Q-Q Plot: Overall IR Memorability",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal()
```



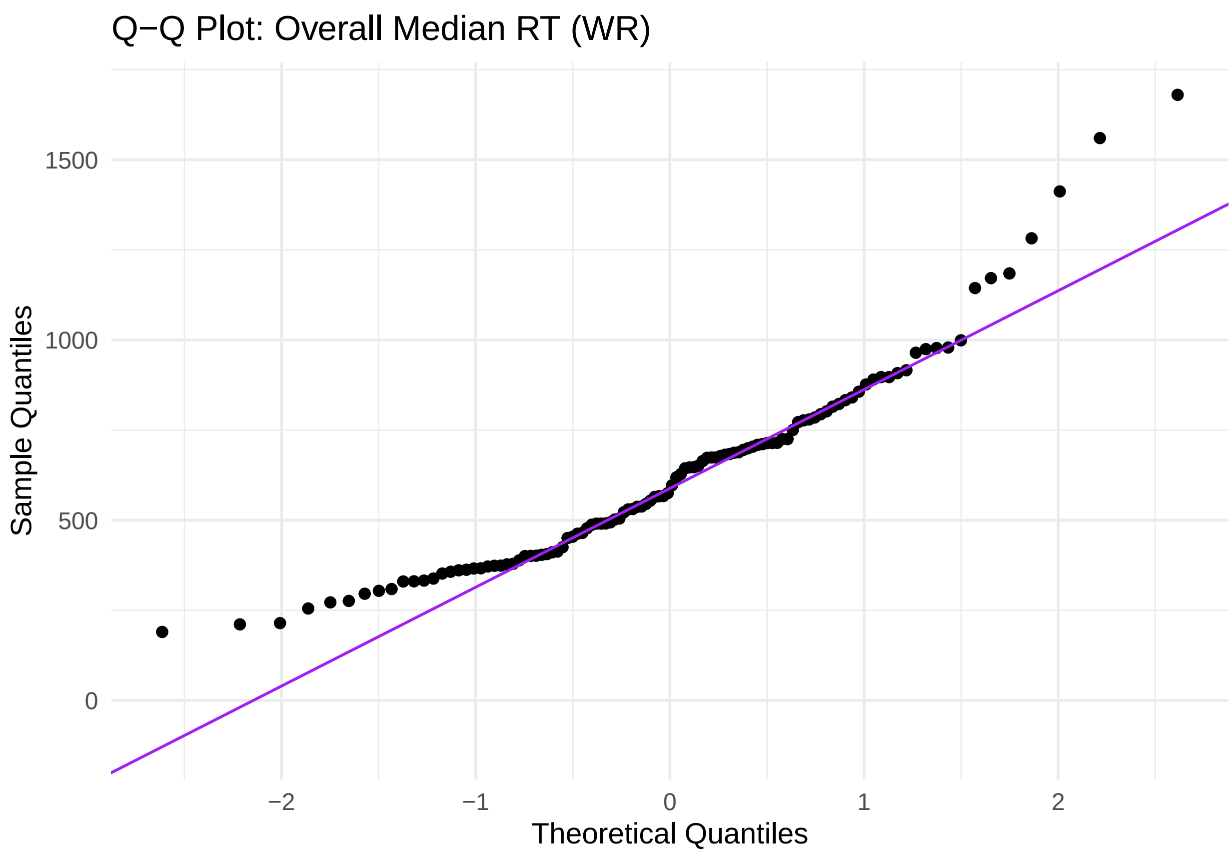

```r
# Q-Q Plot for Overall WR Accuracy
ggplot(h9_data, aes(sample = WR_Accuracy)) +
  stat_qq() +
  stat_qq_line(color = "blue") +
  labs(title = "Q-Q Plot: Overall WR Accuracy",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal()
```




```r
# Q-Q Plot for Overall Median RT (IR)
ggplot(h9_data, aes(sample = Median_RT_IR)) +
  stat_qq() +
  stat_qq_line(color = "darkgreen") +
  labs(title = "Q-Q Plot: Overall Median RT (IR)",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal()
```




```r
# Q-Q Plot for Overall Median RT (WR)
ggplot(h9_data, aes(sample = Median_RT_WR)) +
  stat_qq() +
  stat_qq_line(color = "purple") +
  labs(title = "Q-Q Plot: Overall Median RT (WR)",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal()
```




```r
library(nortest)
library(dplyr)

# Anderson-Darling tests for the specific dataset
overall_normality <- h9_data %>%
  summarise(
    # IR Memorability
    AD_IR_Mem = ad.test(IR_Memorability)$statistic,
    p_val_IR_Mem = ad.test(IR_Memorability)$p.value,

    # WR Accuracy
    AD_WR_Acc = ad.test(WR_Accuracy)$statistic,
    p_val_WR_Acc = ad.test(WR_Accuracy)$p.value,

    # Median Reaction Time (IR)
    AD_RT_IR = ad.test(Median_RT_IR)$statistic,
    p_val_RT_IR = ad.test(Median_RT_IR)$p.value,

    # Median Reaction Time (WR)
    AD_RT_WR = ad.test(Median_RT_WR)$statistic,
    p_val_RT_WR = ad.test(Median_RT_WR)$p.value
  )

# View the results
print(as.data.frame(overall_normality))
```

```
##   AD_IR_Mem p_val_IR_Mem AD_WR_Acc p_val_WR_Acc  AD_RT_IR p_val_RT_IR AD_RT_WR
## 1  2.451727 3.093829e-06 0.3474411    0.4732609 0.2500408   0.7388693 1.714525
##    p_val_RT_WR
## 1 0.0002021714
```

```r
# Using Median RT
cor_ir_median <- cor.test(h9_data$Median_RT_IR, h9_data$IR_Memorability, method = "spearman", exact = FA
print("Spearman Correlation for H9a (IR - Median RT):")
```

```
## [1] "Spearman Correlation for H9a (IR - Median RT):"
```

```r
print(cor_ir_median)
```

```
##
##  Spearman's rank correlation rho
##
## data:  h9_data$Median_RT_IR and h9_data$IR_Memorability
## S = 299595, p-value = 0.002833
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## -0.2795767
```

```r
# Using Mean RT
cor_ir_mean <- cor.test(h9_data$Mean_RT_IR, h9_data$IR_Memorability, method = "spearman", exact = FALSE
print("Spearman Correlation for H9a (IR - Mean RT):")
```

```
## [1] "Spearman Correlation for H9a (IR - Mean RT):"
```

```r
print(cor_ir_mean)
```

```
##
##  Spearman's rank correlation rho
##
## data:  h9_data$Mean_RT_IR and h9_data$IR_Memorability
## S = 302105, p-value = 0.001904
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## -0.2902955
```

```r
# Using Median RT
cor_wr_median <- cor.test(h9_data$Median_RT_WR, h9_data$WR_Accuracy, method = "spearman", exact = FALSE
print("Spearman Correlation for H9b (WR - Median RT):")
```

```
## [1] "Spearman Correlation for H9b (WR - Median RT):"
```

```r
print(cor_wr_median)
```

```
##
##  Spearman's rank correlation rho
##
## data:  h9_data$Median_RT_WR and h9_data$WR_Accuracy
## S = 202452, p-value = 0.1549
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.1353222
```

```r
# Using Mean RT
cor_wr_mean <- cor.test(h9_data$Mean_RT_WR, h9_data$WR_Accuracy, method = "spearman", exact = FALSE)
print("Spearman Correlation for H9b (WR - Mean RT):")
```

```
## [1] "Spearman Correlation for H9b (WR - Mean RT):"
```

```r
print(cor_wr_mean)
```

```
##
##  Spearman's rank correlation rho
##
## data:  h9_data$Mean_RT_WR and h9_data$WR_Accuracy
## S = 207142, p-value = 0.2261
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.1152917
```